

EL9: Documentation

Quarto

Reading

- [1, ch. 28-29] introduces Quarto with good examples

For reference

quarto.org

Levels of documentation

Documentation can exist at several levels in a project.

- **Code level**
 - scripts (comments, e.g. # clean data and merge cohorts)
 - functions (Roxygen2 documentation)
 - notes and reminders (e.g. # TODO:)
- **Project level**
 - README.md
- **Output level**
 - figures and tables
 - reports (PDF / HTML)
 - presentations (slides)

Documenting R scripts

- Section labels by Shift + Ctrl/Cmd + R in Positron and RStudio
- Easiest way to aid navigation in R scripts

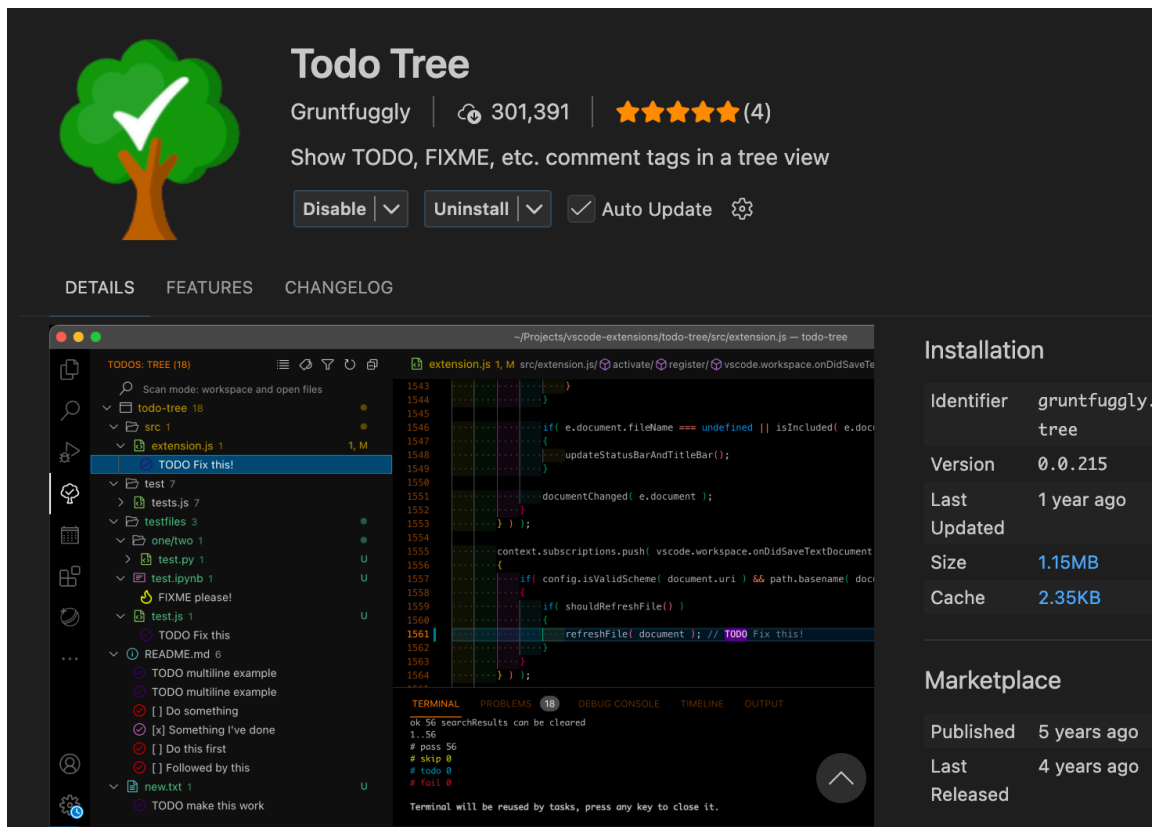
```
# My section heading -----
```

- [commentr package](#)
 - `remotes::install_bitbucket("cancercentrum/commentr")`

```
commentr::header_comment(  
  "My nice script",  
  "Bla bla bla ...",  
  "Erik Bülow",  
  "xxx@yyy.zzz",  
  "John Doe"  
)
```

The comment has been copied to clipboard and can be pasted into a script file!

```
#####  
#                                                                 #  
# Purpose:      My nice script                                  #  
#                                                                 #  
# Author:       Erik Bülow                                     #  
# Contact:      xxx@yyy.zzz                                    #  
# Client:       John Doe                                       #  
#                                                                 #  
# Code created: 2026-03-14                                     #  
# Last updated: 2026-03-14                                     #  
# Source:       /Users/xbuler/STA220/documentation              #  
#                                                                 #  
# Comment:      Bla bla bla ...                                #  
#                                                                 #  
#####
```



Todo Tree
Gruntfuggly | 301,391 | ★★★★★ (4)
Show TODO, FIXME, etc. comment tags in a tree view
Disable | Uninstall | ☒ Auto Update ⚙️

DETAILS | FEATURES | CHANGELOG

Installation

Identifier	gruntfuggly.todo-tree
Version	0.0.215
Last Updated	1 year ago
Size	1.15MB
Cache	2.35KB

Marketplace

Published	5 years ago
Last Released	4 years ago

```
# TODO: Change this to something better:
1 + 2
```

Documenting R functions

What does this function do?

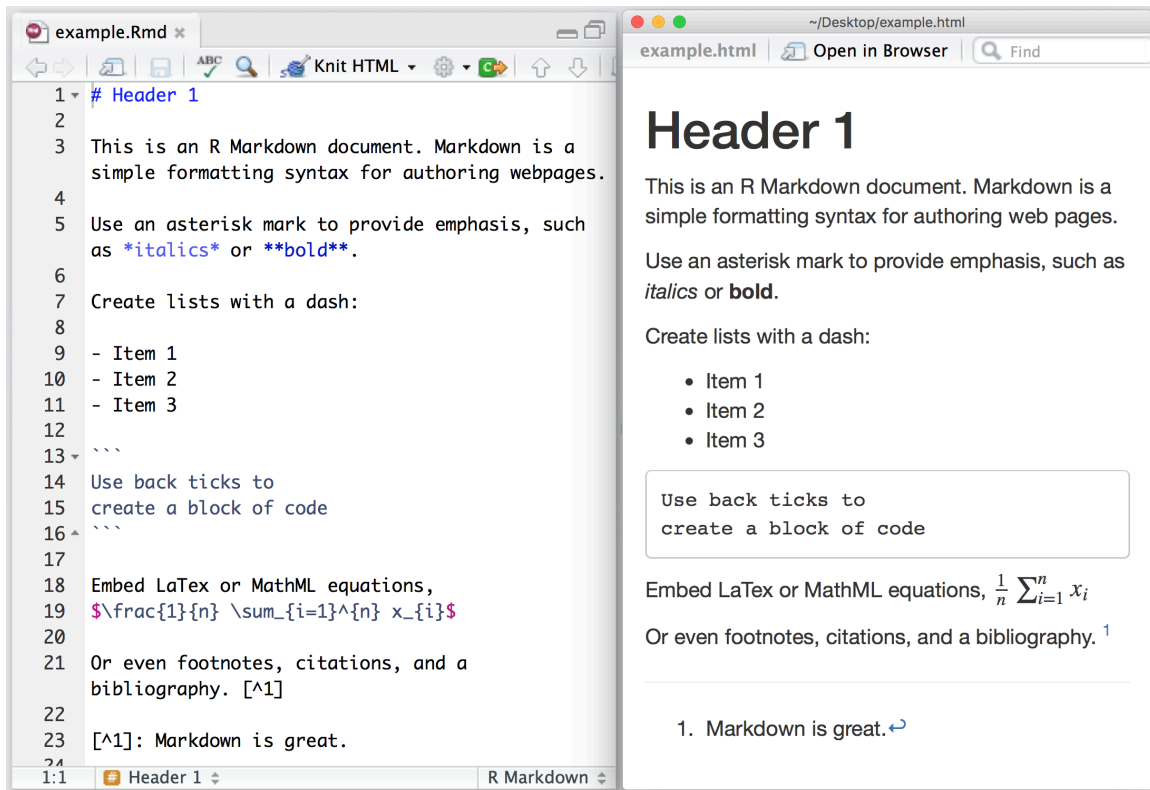
```
hello <- function(who) {
  sprintf("Hello %s!", if (missing(who)) "world" else who)
}
```

```
#' Say hello!
#'
#' This function let's you greet someone or something
#' @param character vector. Someone or something to greet. Will greet the
#' world if argumen is missing.
#' @return character vector of the same length as `character`
#' @examples
#' hello()
#' hello("you")
hello <- function(who) {
  sprintf("Hello %s!", if (missing(who)) "world" else who)
}
```

- There is a standardized syntax to document R functions
- You should be able to click a small light bulb (💡) to insert a sceleton when writing your function
- The same syntax is used within modern R packages
 - but it is useful even within your own projects
- [roxygen2](#) is a package used for package documentation but it also has a great [manual for documenteing functions in general](#)

Documenting a project

- You have a README.md file within your git repository
- It can contain plain text without any formatting
- It will be displayed up-front in GitHub
- You can also include formatting!
- It will be nicely rendered on GitHub or elsewhere



Basic formatting

What you want	Markdown syntax	Result
Heading	# Heading	Heading
Italic	<i>*text*</i>	<i>text</i>
Bold	**text**	text
Inline code	`code`	code
Link	[text](https://example.com)	text
Bullet list	- item	• item
Numbered list	1. item	1. item

i Comment or heading

is used for comments in R files and within R blocks but for headings in .md and .qmd files!

Reproducible Workflows in Biostatistics

Modern biostatistics projects rarely involve only statistics.

- data cleaning and preprocessing
- statistical modelling
- visualisation
- interpretation of results
- reporting and documentation

A good workflow should make it possible to **combine all of these steps in one place**.

What?

Quarto is a publishing system for scientific and technical documents.

It allows you to combine:

- text
- R code
- statistical results
- tables and figures

in a **single workflow**.

This makes it easier to create **reproducible statistical reports**.

Why?

In many projects people work with:

- R scripts for analysis
- Word documents for reporting
- PowerPoint slides for presentations

This separation often leads to:

- copy-paste errors
- outdated figures
- results that cannot easily be reproduced

Quarto helps avoid these problems.

A reproducible analysis document

In a Quarto document you can:

1. write the explanation of your analysis
 2. include the R code used
 3. generate figures and tables automatically
 4. update everything by re-running the document
- If the data or model changes, the **entire report can be regenerated automatically**.
 - It is widely used in **data science, epidemiology and biostatistics**.
 - Integrated in Positron and developed by Posit PBC

Used in three ways

1. For **communicating to decision-makers**, who want to *focus on the conclusions, not the code behind the analysis*.
2. For **collaborating with colleagues** (including future you!), who are interested in both your *conclusions, and how you reached them (i.e. the code)*.
3. As an **environment** in which to do data science and statistics, as a modern-day lab notebook where you can capture not only what you did, but also what you were thinking.

A simple report

- When you chose “New > New File ...” in the upper left corner in Positron you may chose “Quarto document”
- The file will be almost empty but contains:

```
---  
title: "Untitled"  
format: html  
---
```

- This is a YAML header (Yet Another Markup Language)
 - Different formats available:
 - **html**: can also be published online
 - **docx**: for collaborating with non-statisticians etc
 - **typst**: Modern PDF rendering
 - **revealjs**: slide decks (you are looking at one!)
 - Can render to multiple formats at once!
-

This is the yaml header for this presentation:

```
---  
title: "EL9: Documentation"  
subtitle: "Quarto, roxygn2"  
reading: "[@wickham, ch. 28-29]"  
format:  
  revealjs:  
    theme: blood  
    transition: fade  
    slide-number: true  
    css: styles.css  
    smaller: true  
    scrollable: true  
    incremental: false  
  typst:  
    include-before-body:  
      - text: |
```

```

// Gör alla länkar tydliga i PDF: blå + understrukna
#show link: set text(fill: rgb("1a5fb4"))
#show link: underline
docx: default
bibliography: references.bib
---
```

YAML structure

YAML uses **indentation to represent structure**.

Think of it like a tree.

```

format:
  html:
    code-fold: true
```

This corresponds to the structure:

```

format
├─ html
│   └─ code-fold: true
```

What happens if indentation is wrong?

```

format:
html:
  code-fold: true
```

Now Quarto interprets this as:

```

format
html
├─ code-fold: true
```

This is **not the intended structure**, and the document may fail to render.

💡 Rule of thumb

- use **two spaces per level**
- never use **tabs**
- when something fails, **check indentation first**

Three parts

Quarto documents have three parts:

1. An (optional) YAML header surrounded by ---s.
2. Chunks of R code surrounded by ```{r} and ```.

```
```{r}          ← chunk header  
#| echo: false ← options controlling the output
summary(x) ← R code
```            ← end of chunk
```

3. Text with markdown format

Where 1 (YAML header) comes first and where 2 and 3 are intermingled throughout the rest of the document.

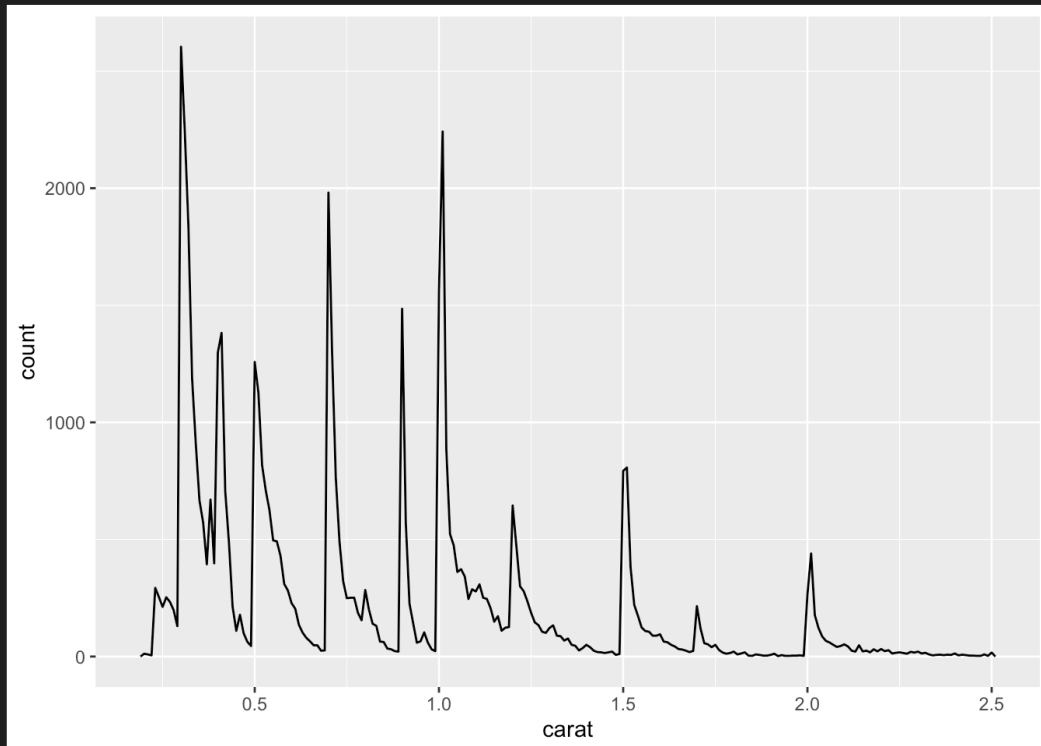
```
Preview ☐ Render on Save 🔍 📄 📌 ...
```

```
1 ---
2 title: "Diamond sizes"
3 date: 2022-09-12
4 format: html
5 ---
6
7 >Run Cell | Run Next Cell
8 ```{r}
9 #| label: setup
10 #| include: false
11
12 library(tidyverse)
13
14 smaller <- diamonds |>
15   filter(carat <= 2.5)
16   ```
17
18 We have data about `r nrow(diamonds)` diamonds.
19 Only `r nrow(diamonds) - nrow(smaller)` are larger than 2.5 carats.
20 The distribution of the remainder is shown below:
21
22 >Run Cell | Run Above
23 ```{r}
24 #| label: plot-smaller-diamonds
25 #| echo: false
26
27 smaller |>
28   ggplot(aes(x = carat)) +
29     geom_freqpoly(binwidth = 0.01)
30   ```
```

Diamond sizes

PUBLISHED
September 12, 2022

We have data about 53940 diamonds. Only 126 are larger than 2.5 carats. The distribution of the remainder is shown below:



Inline code

Sometimes we want to insert **small pieces of R output directly in the text**.

This is called **inline code**.

Inline code will always just show the result of the call (not the code itself in the rendered document):

```
Todays date is `r Sys.Date()` and it is `r format(Sys.time(), format = "%H")`
```

Todys date is 2026-03-14 and it is 06 PM

To highlight with bold and italic:

```
Todays date is **`r Sys.Date()`** and it is *`r format(Sys.time(), format = "%I %p")`*
```

Todays date is **2026-03-14** and it is *06 PM*

To publish

Decision makers, stake holders, friends or the public might not care about R. Disable R code (echoing the code itself), warnings and messages:

```
---  
title: "Untitled"  
format: html  
execute:  
  echo: false  
  warning: false  
  message: false  
---
```

For collaborators

- Might still care less about verbose R messages and warnings (they trust that you have already considered those)
- May check the underlying R code if relevant (use code-fold)

```
---  
title: "Untitled"  
format:  
  html:  
    code-fold: true  
execute:  
  warning: false  
  message: false  
---
```

This can also be set individually for each code chunk:

```
```{r}  
#| echo: false
#| warning: false
#| message: true
message("will be muted!")
warning("will be ignored")
plot(something_beautiful, reayl = TRUE)
```
```

Figures

Figures are usually displayed without problem (and you can adjust their size, add captions etc):

```
```{r}
#| fig-cap: This is origo!
plot(0, 0)
```
```

```
plot(0, 0)
```

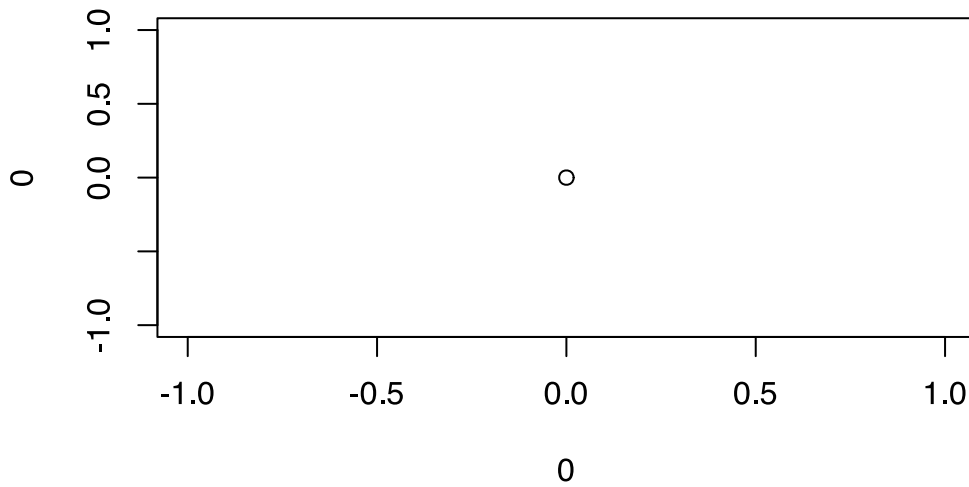


Figure 1: This is origo!

Tables

Just printing the R output might not look very nice:

```
iris
```

| | Sepal.Length | Sepal.Width | Petal.Length | Petal.Width | Species |
|---|--------------|-------------|--------------|-------------|---------|
| 1 | 5.1 | 3.5 | 1.4 | 0.2 | setosa |
| 2 | 4.9 | 3.0 | 1.4 | 0.2 | setosa |
| 3 | 4.7 | 3.2 | 1.3 | 0.2 | setosa |
| 4 | 4.6 | 3.1 | 1.5 | 0.2 | setosa |
| 5 | 5.0 | 3.6 | 1.4 | 0.2 | setosa |
| 6 | 5.4 | 3.9 | 1.7 | 0.4 | setosa |
| 7 | 4.6 | 3.4 | 1.4 | 0.3 | setosa |

| | | | | | |
|----|-----|-----|-----|-----|------------|
| 8 | 5.0 | 3.4 | 1.5 | 0.2 | setosa |
| 9 | 4.4 | 2.9 | 1.4 | 0.2 | setosa |
| 10 | 4.9 | 3.1 | 1.5 | 0.1 | setosa |
| 11 | 5.4 | 3.7 | 1.5 | 0.2 | setosa |
| 12 | 4.8 | 3.4 | 1.6 | 0.2 | setosa |
| 13 | 4.8 | 3.0 | 1.4 | 0.1 | setosa |
| 14 | 4.3 | 3.0 | 1.1 | 0.1 | setosa |
| 15 | 5.8 | 4.0 | 1.2 | 0.2 | setosa |
| 16 | 5.7 | 4.4 | 1.5 | 0.4 | setosa |
| 17 | 5.4 | 3.9 | 1.3 | 0.4 | setosa |
| 18 | 5.1 | 3.5 | 1.4 | 0.3 | setosa |
| 19 | 5.7 | 3.8 | 1.7 | 0.3 | setosa |
| 20 | 5.1 | 3.8 | 1.5 | 0.3 | setosa |
| 21 | 5.4 | 3.4 | 1.7 | 0.2 | setosa |
| 22 | 5.1 | 3.7 | 1.5 | 0.4 | setosa |
| 23 | 4.6 | 3.6 | 1.0 | 0.2 | setosa |
| 24 | 5.1 | 3.3 | 1.7 | 0.5 | setosa |
| 25 | 4.8 | 3.4 | 1.9 | 0.2 | setosa |
| 26 | 5.0 | 3.0 | 1.6 | 0.2 | setosa |
| 27 | 5.0 | 3.4 | 1.6 | 0.4 | setosa |
| 28 | 5.2 | 3.5 | 1.5 | 0.2 | setosa |
| 29 | 5.2 | 3.4 | 1.4 | 0.2 | setosa |
| 30 | 4.7 | 3.2 | 1.6 | 0.2 | setosa |
| 31 | 4.8 | 3.1 | 1.6 | 0.2 | setosa |
| 32 | 5.4 | 3.4 | 1.5 | 0.4 | setosa |
| 33 | 5.2 | 4.1 | 1.5 | 0.1 | setosa |
| 34 | 5.5 | 4.2 | 1.4 | 0.2 | setosa |
| 35 | 4.9 | 3.1 | 1.5 | 0.2 | setosa |
| 36 | 5.0 | 3.2 | 1.2 | 0.2 | setosa |
| 37 | 5.5 | 3.5 | 1.3 | 0.2 | setosa |
| 38 | 4.9 | 3.6 | 1.4 | 0.1 | setosa |
| 39 | 4.4 | 3.0 | 1.3 | 0.2 | setosa |
| 40 | 5.1 | 3.4 | 1.5 | 0.2 | setosa |
| 41 | 5.0 | 3.5 | 1.3 | 0.3 | setosa |
| 42 | 4.5 | 2.3 | 1.3 | 0.3 | setosa |
| 43 | 4.4 | 3.2 | 1.3 | 0.2 | setosa |
| 44 | 5.0 | 3.5 | 1.6 | 0.6 | setosa |
| 45 | 5.1 | 3.8 | 1.9 | 0.4 | setosa |
| 46 | 4.8 | 3.0 | 1.4 | 0.3 | setosa |
| 47 | 5.1 | 3.8 | 1.6 | 0.2 | setosa |
| 48 | 4.6 | 3.2 | 1.4 | 0.2 | setosa |
| 49 | 5.3 | 3.7 | 1.5 | 0.2 | setosa |
| 50 | 5.0 | 3.3 | 1.4 | 0.2 | setosa |
| 51 | 7.0 | 3.2 | 4.7 | 1.4 | versicolor |
| 52 | 6.4 | 3.2 | 4.5 | 1.5 | versicolor |
| 53 | 6.9 | 3.1 | 4.9 | 1.5 | versicolor |
| 54 | 5.5 | 2.3 | 4.0 | 1.3 | versicolor |
| 55 | 6.5 | 2.8 | 4.6 | 1.5 | versicolor |

| | | | | |
|-----|-----|-----|-----|----------------|
| 56 | 5.7 | 2.8 | 4.5 | 1.3 versicolor |
| 57 | 6.3 | 3.3 | 4.7 | 1.6 versicolor |
| 58 | 4.9 | 2.4 | 3.3 | 1.0 versicolor |
| 59 | 6.6 | 2.9 | 4.6 | 1.3 versicolor |
| 60 | 5.2 | 2.7 | 3.9 | 1.4 versicolor |
| 61 | 5.0 | 2.0 | 3.5 | 1.0 versicolor |
| 62 | 5.9 | 3.0 | 4.2 | 1.5 versicolor |
| 63 | 6.0 | 2.2 | 4.0 | 1.0 versicolor |
| 64 | 6.1 | 2.9 | 4.7 | 1.4 versicolor |
| 65 | 5.6 | 2.9 | 3.6 | 1.3 versicolor |
| 66 | 6.7 | 3.1 | 4.4 | 1.4 versicolor |
| 67 | 5.6 | 3.0 | 4.5 | 1.5 versicolor |
| 68 | 5.8 | 2.7 | 4.1 | 1.0 versicolor |
| 69 | 6.2 | 2.2 | 4.5 | 1.5 versicolor |
| 70 | 5.6 | 2.5 | 3.9 | 1.1 versicolor |
| 71 | 5.9 | 3.2 | 4.8 | 1.8 versicolor |
| 72 | 6.1 | 2.8 | 4.0 | 1.3 versicolor |
| 73 | 6.3 | 2.5 | 4.9 | 1.5 versicolor |
| 74 | 6.1 | 2.8 | 4.7 | 1.2 versicolor |
| 75 | 6.4 | 2.9 | 4.3 | 1.3 versicolor |
| 76 | 6.6 | 3.0 | 4.4 | 1.4 versicolor |
| 77 | 6.8 | 2.8 | 4.8 | 1.4 versicolor |
| 78 | 6.7 | 3.0 | 5.0 | 1.7 versicolor |
| 79 | 6.0 | 2.9 | 4.5 | 1.5 versicolor |
| 80 | 5.7 | 2.6 | 3.5 | 1.0 versicolor |
| 81 | 5.5 | 2.4 | 3.8 | 1.1 versicolor |
| 82 | 5.5 | 2.4 | 3.7 | 1.0 versicolor |
| 83 | 5.8 | 2.7 | 3.9 | 1.2 versicolor |
| 84 | 6.0 | 2.7 | 5.1 | 1.6 versicolor |
| 85 | 5.4 | 3.0 | 4.5 | 1.5 versicolor |
| 86 | 6.0 | 3.4 | 4.5 | 1.6 versicolor |
| 87 | 6.7 | 3.1 | 4.7 | 1.5 versicolor |
| 88 | 6.3 | 2.3 | 4.4 | 1.3 versicolor |
| 89 | 5.6 | 3.0 | 4.1 | 1.3 versicolor |
| 90 | 5.5 | 2.5 | 4.0 | 1.3 versicolor |
| 91 | 5.5 | 2.6 | 4.4 | 1.2 versicolor |
| 92 | 6.1 | 3.0 | 4.6 | 1.4 versicolor |
| 93 | 5.8 | 2.6 | 4.0 | 1.2 versicolor |
| 94 | 5.0 | 2.3 | 3.3 | 1.0 versicolor |
| 95 | 5.6 | 2.7 | 4.2 | 1.3 versicolor |
| 96 | 5.7 | 3.0 | 4.2 | 1.2 versicolor |
| 97 | 5.7 | 2.9 | 4.2 | 1.3 versicolor |
| 98 | 6.2 | 2.9 | 4.3 | 1.3 versicolor |
| 99 | 5.1 | 2.5 | 3.0 | 1.1 versicolor |
| 100 | 5.7 | 2.8 | 4.1 | 1.3 versicolor |
| 101 | 6.3 | 3.3 | 6.0 | 2.5 virginica |
| 102 | 5.8 | 2.7 | 5.1 | 1.9 virginica |
| 103 | 7.1 | 3.0 | 5.9 | 2.1 virginica |

| | | | | | |
|-----|-----|-----|-----|-----|-----------|
| 104 | 6.3 | 2.9 | 5.6 | 1.8 | virginica |
| 105 | 6.5 | 3.0 | 5.8 | 2.2 | virginica |
| 106 | 7.6 | 3.0 | 6.6 | 2.1 | virginica |
| 107 | 4.9 | 2.5 | 4.5 | 1.7 | virginica |
| 108 | 7.3 | 2.9 | 6.3 | 1.8 | virginica |
| 109 | 6.7 | 2.5 | 5.8 | 1.8 | virginica |
| 110 | 7.2 | 3.6 | 6.1 | 2.5 | virginica |
| 111 | 6.5 | 3.2 | 5.1 | 2.0 | virginica |
| 112 | 6.4 | 2.7 | 5.3 | 1.9 | virginica |
| 113 | 6.8 | 3.0 | 5.5 | 2.1 | virginica |
| 114 | 5.7 | 2.5 | 5.0 | 2.0 | virginica |
| 115 | 5.8 | 2.8 | 5.1 | 2.4 | virginica |
| 116 | 6.4 | 3.2 | 5.3 | 2.3 | virginica |
| 117 | 6.5 | 3.0 | 5.5 | 1.8 | virginica |
| 118 | 7.7 | 3.8 | 6.7 | 2.2 | virginica |
| 119 | 7.7 | 2.6 | 6.9 | 2.3 | virginica |
| 120 | 6.0 | 2.2 | 5.0 | 1.5 | virginica |
| 121 | 6.9 | 3.2 | 5.7 | 2.3 | virginica |
| 122 | 5.6 | 2.8 | 4.9 | 2.0 | virginica |
| 123 | 7.7 | 2.8 | 6.7 | 2.0 | virginica |
| 124 | 6.3 | 2.7 | 4.9 | 1.8 | virginica |
| 125 | 6.7 | 3.3 | 5.7 | 2.1 | virginica |
| 126 | 7.2 | 3.2 | 6.0 | 1.8 | virginica |
| 127 | 6.2 | 2.8 | 4.8 | 1.8 | virginica |
| 128 | 6.1 | 3.0 | 4.9 | 1.8 | virginica |
| 129 | 6.4 | 2.8 | 5.6 | 2.1 | virginica |
| 130 | 7.2 | 3.0 | 5.8 | 1.6 | virginica |
| 131 | 7.4 | 2.8 | 6.1 | 1.9 | virginica |
| 132 | 7.9 | 3.8 | 6.4 | 2.0 | virginica |
| 133 | 6.4 | 2.8 | 5.6 | 2.2 | virginica |
| 134 | 6.3 | 2.8 | 5.1 | 1.5 | virginica |
| 135 | 6.1 | 2.6 | 5.6 | 1.4 | virginica |
| 136 | 7.7 | 3.0 | 6.1 | 2.3 | virginica |
| 137 | 6.3 | 3.4 | 5.6 | 2.4 | virginica |
| 138 | 6.4 | 3.1 | 5.5 | 1.8 | virginica |
| 139 | 6.0 | 3.0 | 4.8 | 1.8 | virginica |
| 140 | 6.9 | 3.1 | 5.4 | 2.1 | virginica |
| 141 | 6.7 | 3.1 | 5.6 | 2.4 | virginica |
| 142 | 6.9 | 3.1 | 5.1 | 2.3 | virginica |
| 143 | 5.8 | 2.7 | 5.1 | 1.9 | virginica |
| 144 | 6.8 | 3.2 | 5.9 | 2.3 | virginica |
| 145 | 6.7 | 3.3 | 5.7 | 2.5 | virginica |
| 146 | 6.7 | 3.0 | 5.2 | 2.3 | virginica |
| 147 | 6.3 | 2.5 | 5.0 | 1.9 | virginica |
| 148 | 6.5 | 3.0 | 5.2 | 2.0 | virginica |
| 149 | 6.2 | 3.4 | 5.4 | 2.3 | virginica |
| 150 | 5.9 | 3.0 | 5.1 | 1.8 | virginica |

Better with kable

The easiest way to make tibbles/data.frames/data.tables look nicer is with `df-print: kable` in the yaml header.

```
---
title: "Untitled"
format:
  html:
    df-print: kable
---
```

```
knitr::kable(iris)
```

| Sepal.Length | Sepal.Width | Petal.Length | Petal.Width | Species |
|--------------|-------------|--------------|-------------|---------|
| 5.1 | 3.5 | 1.4 | 0.2 | setosa |
| 4.9 | 3.0 | 1.4 | 0.2 | setosa |
| 4.7 | 3.2 | 1.3 | 0.2 | setosa |
| 4.6 | 3.1 | 1.5 | 0.2 | setosa |
| 5.0 | 3.6 | 1.4 | 0.2 | setosa |
| 5.4 | 3.9 | 1.7 | 0.4 | setosa |
| 4.6 | 3.4 | 1.4 | 0.3 | setosa |
| 5.0 | 3.4 | 1.5 | 0.2 | setosa |
| 4.4 | 2.9 | 1.4 | 0.2 | setosa |
| 4.9 | 3.1 | 1.5 | 0.1 | setosa |
| 5.4 | 3.7 | 1.5 | 0.2 | setosa |
| 4.8 | 3.4 | 1.6 | 0.2 | setosa |
| 4.8 | 3.0 | 1.4 | 0.1 | setosa |
| 4.3 | 3.0 | 1.1 | 0.1 | setosa |
| 5.8 | 4.0 | 1.2 | 0.2 | setosa |
| 5.7 | 4.4 | 1.5 | 0.4 | setosa |
| 5.4 | 3.9 | 1.3 | 0.4 | setosa |
| 5.1 | 3.5 | 1.4 | 0.3 | setosa |
| 5.7 | 3.8 | 1.7 | 0.3 | setosa |
| 5.1 | 3.8 | 1.5 | 0.3 | setosa |
| 5.4 | 3.4 | 1.7 | 0.2 | setosa |

| Sepal.Length | Sepal.Width | Petal.Length | Petal.Width | Species |
|--------------|-------------|--------------|-------------|------------|
| 5.1 | 3.7 | 1.5 | 0.4 | setosa |
| 4.6 | 3.6 | 1.0 | 0.2 | setosa |
| 5.1 | 3.3 | 1.7 | 0.5 | setosa |
| 4.8 | 3.4 | 1.9 | 0.2 | setosa |
| 5.0 | 3.0 | 1.6 | 0.2 | setosa |
| 5.0 | 3.4 | 1.6 | 0.4 | setosa |
| 5.2 | 3.5 | 1.5 | 0.2 | setosa |
| 5.2 | 3.4 | 1.4 | 0.2 | setosa |
| 4.7 | 3.2 | 1.6 | 0.2 | setosa |
| 4.8 | 3.1 | 1.6 | 0.2 | setosa |
| 5.4 | 3.4 | 1.5 | 0.4 | setosa |
| 5.2 | 4.1 | 1.5 | 0.1 | setosa |
| 5.5 | 4.2 | 1.4 | 0.2 | setosa |
| 4.9 | 3.1 | 1.5 | 0.2 | setosa |
| 5.0 | 3.2 | 1.2 | 0.2 | setosa |
| 5.5 | 3.5 | 1.3 | 0.2 | setosa |
| 4.9 | 3.6 | 1.4 | 0.1 | setosa |
| 4.4 | 3.0 | 1.3 | 0.2 | setosa |
| 5.1 | 3.4 | 1.5 | 0.2 | setosa |
| 5.0 | 3.5 | 1.3 | 0.3 | setosa |
| 4.5 | 2.3 | 1.3 | 0.3 | setosa |
| 4.4 | 3.2 | 1.3 | 0.2 | setosa |
| 5.0 | 3.5 | 1.6 | 0.6 | setosa |
| 5.1 | 3.8 | 1.9 | 0.4 | setosa |
| 4.8 | 3.0 | 1.4 | 0.3 | setosa |
| 5.1 | 3.8 | 1.6 | 0.2 | setosa |
| 4.6 | 3.2 | 1.4 | 0.2 | setosa |
| 5.3 | 3.7 | 1.5 | 0.2 | setosa |
| 5.0 | 3.3 | 1.4 | 0.2 | setosa |
| 7.0 | 3.2 | 4.7 | 1.4 | versicolor |

| Sepal.Length | Sepal.Width | Petal.Length | Petal.Width | Species |
|--------------|-------------|--------------|-------------|------------|
| 6.4 | 3.2 | 4.5 | 1.5 | versicolor |
| 6.9 | 3.1 | 4.9 | 1.5 | versicolor |
| 5.5 | 2.3 | 4.0 | 1.3 | versicolor |
| 6.5 | 2.8 | 4.6 | 1.5 | versicolor |
| 5.7 | 2.8 | 4.5 | 1.3 | versicolor |
| 6.3 | 3.3 | 4.7 | 1.6 | versicolor |
| 4.9 | 2.4 | 3.3 | 1.0 | versicolor |
| 6.6 | 2.9 | 4.6 | 1.3 | versicolor |
| 5.2 | 2.7 | 3.9 | 1.4 | versicolor |
| 5.0 | 2.0 | 3.5 | 1.0 | versicolor |
| 5.9 | 3.0 | 4.2 | 1.5 | versicolor |
| 6.0 | 2.2 | 4.0 | 1.0 | versicolor |
| 6.1 | 2.9 | 4.7 | 1.4 | versicolor |
| 5.6 | 2.9 | 3.6 | 1.3 | versicolor |
| 6.7 | 3.1 | 4.4 | 1.4 | versicolor |
| 5.6 | 3.0 | 4.5 | 1.5 | versicolor |
| 5.8 | 2.7 | 4.1 | 1.0 | versicolor |
| 6.2 | 2.2 | 4.5 | 1.5 | versicolor |
| 5.6 | 2.5 | 3.9 | 1.1 | versicolor |
| 5.9 | 3.2 | 4.8 | 1.8 | versicolor |
| 6.1 | 2.8 | 4.0 | 1.3 | versicolor |
| 6.3 | 2.5 | 4.9 | 1.5 | versicolor |
| 6.1 | 2.8 | 4.7 | 1.2 | versicolor |
| 6.4 | 2.9 | 4.3 | 1.3 | versicolor |
| 6.6 | 3.0 | 4.4 | 1.4 | versicolor |
| 6.8 | 2.8 | 4.8 | 1.4 | versicolor |
| 6.7 | 3.0 | 5.0 | 1.7 | versicolor |
| 6.0 | 2.9 | 4.5 | 1.5 | versicolor |
| 5.7 | 2.6 | 3.5 | 1.0 | versicolor |
| 5.5 | 2.4 | 3.8 | 1.1 | versicolor |

| Sepal.Length | Sepal.Width | Petal.Length | Petal.Width | Species |
|--------------|-------------|--------------|-------------|------------|
| 5.5 | 2.4 | 3.7 | 1.0 | versicolor |
| 5.8 | 2.7 | 3.9 | 1.2 | versicolor |
| 6.0 | 2.7 | 5.1 | 1.6 | versicolor |
| 5.4 | 3.0 | 4.5 | 1.5 | versicolor |
| 6.0 | 3.4 | 4.5 | 1.6 | versicolor |
| 6.7 | 3.1 | 4.7 | 1.5 | versicolor |
| 6.3 | 2.3 | 4.4 | 1.3 | versicolor |
| 5.6 | 3.0 | 4.1 | 1.3 | versicolor |
| 5.5 | 2.5 | 4.0 | 1.3 | versicolor |
| 5.5 | 2.6 | 4.4 | 1.2 | versicolor |
| 6.1 | 3.0 | 4.6 | 1.4 | versicolor |
| 5.8 | 2.6 | 4.0 | 1.2 | versicolor |
| 5.0 | 2.3 | 3.3 | 1.0 | versicolor |
| 5.6 | 2.7 | 4.2 | 1.3 | versicolor |
| 5.7 | 3.0 | 4.2 | 1.2 | versicolor |
| 5.7 | 2.9 | 4.2 | 1.3 | versicolor |
| 6.2 | 2.9 | 4.3 | 1.3 | versicolor |
| 5.1 | 2.5 | 3.0 | 1.1 | versicolor |
| 5.7 | 2.8 | 4.1 | 1.3 | versicolor |
| 6.3 | 3.3 | 6.0 | 2.5 | virginica |
| 5.8 | 2.7 | 5.1 | 1.9 | virginica |
| 7.1 | 3.0 | 5.9 | 2.1 | virginica |
| 6.3 | 2.9 | 5.6 | 1.8 | virginica |
| 6.5 | 3.0 | 5.8 | 2.2 | virginica |
| 7.6 | 3.0 | 6.6 | 2.1 | virginica |
| 4.9 | 2.5 | 4.5 | 1.7 | virginica |
| 7.3 | 2.9 | 6.3 | 1.8 | virginica |
| 6.7 | 2.5 | 5.8 | 1.8 | virginica |
| 7.2 | 3.6 | 6.1 | 2.5 | virginica |
| 6.5 | 3.2 | 5.1 | 2.0 | virginica |

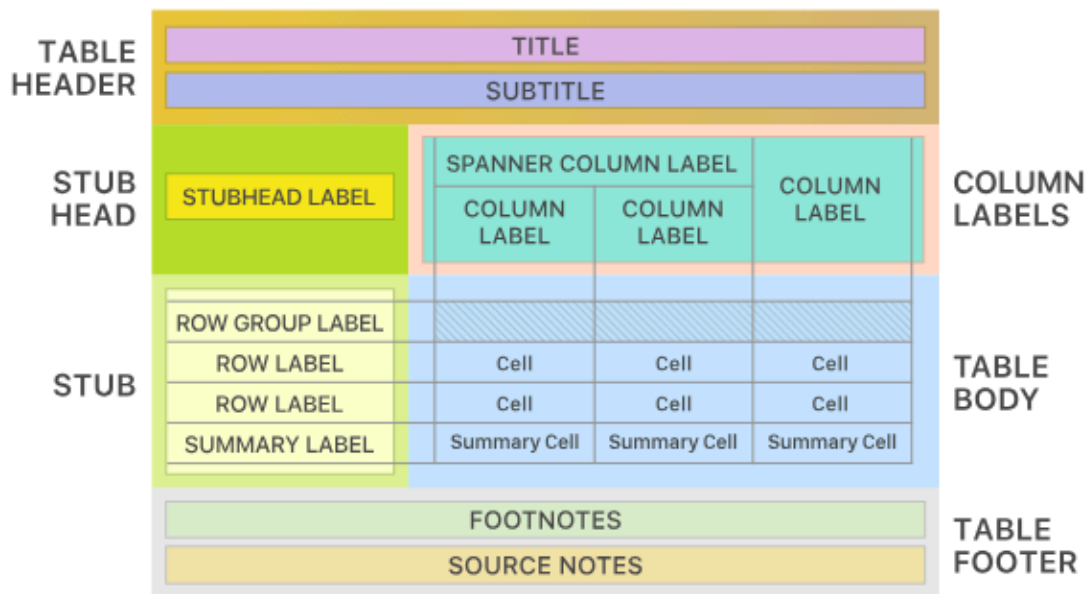
| Sepal.Length | Sepal.Width | Petal.Length | Petal.Width | Species |
|--------------|-------------|--------------|-------------|-----------|
| 6.4 | 2.7 | 5.3 | 1.9 | virginica |
| 6.8 | 3.0 | 5.5 | 2.1 | virginica |
| 5.7 | 2.5 | 5.0 | 2.0 | virginica |
| 5.8 | 2.8 | 5.1 | 2.4 | virginica |
| 6.4 | 3.2 | 5.3 | 2.3 | virginica |
| 6.5 | 3.0 | 5.5 | 1.8 | virginica |
| 7.7 | 3.8 | 6.7 | 2.2 | virginica |
| 7.7 | 2.6 | 6.9 | 2.3 | virginica |
| 6.0 | 2.2 | 5.0 | 1.5 | virginica |
| 6.9 | 3.2 | 5.7 | 2.3 | virginica |
| 5.6 | 2.8 | 4.9 | 2.0 | virginica |
| 7.7 | 2.8 | 6.7 | 2.0 | virginica |
| 6.3 | 2.7 | 4.9 | 1.8 | virginica |
| 6.7 | 3.3 | 5.7 | 2.1 | virginica |
| 7.2 | 3.2 | 6.0 | 1.8 | virginica |
| 6.2 | 2.8 | 4.8 | 1.8 | virginica |
| 6.1 | 3.0 | 4.9 | 1.8 | virginica |
| 6.4 | 2.8 | 5.6 | 2.1 | virginica |
| 7.2 | 3.0 | 5.8 | 1.6 | virginica |
| 7.4 | 2.8 | 6.1 | 1.9 | virginica |
| 7.9 | 3.8 | 6.4 | 2.0 | virginica |
| 6.4 | 2.8 | 5.6 | 2.2 | virginica |
| 6.3 | 2.8 | 5.1 | 1.5 | virginica |
| 6.1 | 2.6 | 5.6 | 1.4 | virginica |
| 7.7 | 3.0 | 6.1 | 2.3 | virginica |
| 6.3 | 3.4 | 5.6 | 2.4 | virginica |
| 6.4 | 3.1 | 5.5 | 1.8 | virginica |
| 6.0 | 3.0 | 4.8 | 1.8 | virginica |
| 6.9 | 3.1 | 5.4 | 2.1 | virginica |
| 6.7 | 3.1 | 5.6 | 2.4 | virginica |

| Sepal.Length | Sepal.Width | Petal.Length | Petal.Width | Species |
|--------------|-------------|--------------|-------------|-----------|
| 6.9 | 3.1 | 5.1 | 2.3 | virginica |
| 5.8 | 2.7 | 5.1 | 1.9 | virginica |
| 6.8 | 3.2 | 5.9 | 2.3 | virginica |
| 6.7 | 3.3 | 5.7 | 2.5 | virginica |
| 6.7 | 3.0 | 5.2 | 2.3 | virginica |
| 6.3 | 2.5 | 5.0 | 1.9 | virginica |
| 6.5 | 3.0 | 5.2 | 2.0 | virginica |
| 6.2 | 3.4 | 5.4 | 2.3 | virginica |
| 5.9 | 3.0 | 5.1 | 1.8 | virginica |

Best with gt

- Very flexible
- Publication ready
- [Package from Posit](#)
- Sometimes [flextable](#) if rendering MS Office documents (Word, PowerPoint)

The Parts of a gt Table



A Typical gt Workflow



gtsummary

[gtsummary package](#) build on top of {gt} to easily summarise datasets, regression models etc

```
```{r}
gtsummary::tbl_summary(iris, Species)
```
```

```
gtsummary::tbl_summary(iris, Species)
```

| | setosa | versicolor | virginica |
|----------------|-------------------|-------------------|-------------------|
| Characteristic | N = 50' | N = 50' | N = 50' |
| Sepal.Length | 5.00 (4.80, 5.20) | 5.90 (5.60, 6.30) | 6.50 (6.20, 6.90) |
| Sepal.Width | 3.40 (3.20, 3.70) | 2.80 (2.50, 3.00) | 3.00 (2.80, 3.20) |
| Petal.Length | 1.50 (1.40, 1.60) | 4.35 (4.00, 4.60) | 5.55 (5.10, 5.90) |
| Petal.Width | 0.20 (0.20, 0.30) | 1.30 (1.20, 1.50) | 2.00 (1.80, 2.30) |

' Median (Q1, Q3)

Presentations

- Quarto can export to PowerPoint
- And to the older Beamer format
- But I recommend [revealjs](#)
 - format: revealjs
 - can also be modified with speaker notes, laser pointer etc
- [Info and examples](#)

For examplen (as used for those slides):

```
format:
  revealjs:
    theme: blood
    transition: fade
    slide-number: true
    css: styles.css
    smaller: true
    scrollable: true
    incremental: false
```

Manuscripts

- [Quarto Manuscript](#)
- framework for writing and publishing scholarly articles
- Something for your later thesis work?
- Handles references
- Export to Word documents (.docx) etc
 - which is still the most common format for submission to medical/epidemiological journals
 - Still used for collaborative work with non-statisticians etc

Use references

When writing scientific reports we need to:

- acknowledge previous research
- support claims with evidence
- allow readers to find the original sources

This is done through **citations** and a **reference list**.

Example:

Don't blame me, it was according to H. Wickham, M. Çetinkaya-Rundel, and G. Grolemund [1].

At the end of the document a **bibliography** is automatically generated.

What is reference management?

Reference management means keeping track of:

- articles, books, reports, datasets

including information such as:

- authors, title, journal, year
- [DOI](#)

Instead of typing references manually, software can **store and format them automatically**.

Common reference management tools

Many researchers use tools such as:

- **Zotero**
- **Mendeley**
- **EndNote**

These tools allow you to:

- collect references from databases (PubMed, Google Scholar, etc.)
- organise them in libraries
- export them to writing tools

Quarto can work with references from these systems directly.

How Quarto handles references

Quarto uses a **bibliography file**, usually in `.bib` format.

Example in the document header:

```
bibliography: references.bib
```

Then citations can be written directly in the text:

```
Several studies have examined this relationship @smith2020.
```

Quarto will render something like:

Several studies have examined this relationship (Smith 2020).

The full reference will appear automatically in the bibliography.

Integrate with {targets}

[tarchetypes::tar_quarto\(\)](#)

in_targets.R

```
library(targets)
library(tarchetypes) # Let you integrate a quarto report/presentation
list(
  tar_target(data, data.frame(x = seq_len(26), y = letters))
  tar_quarto(report, "report.qmd") # Make presentation
)
```


In report.qmd

```
---
title: My report
format: html
---

# Here is my beautiful data!

```{r}
gt::gt(tar_read(data))
```
```

Try it

Copy this code to the terminal (not the console!) and try `targets::tar_make()` in the console!

```
cd
mkdir test_targets_quarto
cat > test_targets_quarto/_targets.R <<'EOF'
library(targets)
library(tarchetypes) # Lets you integrate a quarto report/presentation

list(
  tar_target(data, iris),
  tar_target(model, lm(Sepal.Length ~ ., data)),
  tar_quarto(report, "report.qmd") # Make presentation
)
EOF

cat > test_targets_quarto/report.qmd <<'EOF'
---
title: My project
author: My name
date: today
format:
  typst: default
  docx: default
  html: default
  revealjs:
    output-file: slides.html
---

## Here are some flowers!

```{r}
gtsummary::tbl_summary(targets::tar_read(data))
```
```

```
## And here is a model

```{r}
gtsummary::tbl_regression(targets::tar_read(model))
```

EOF

positron test_targets_quarto
# rm -r test_targets_quarto
```

Online portfolio

- Looking for a job after you finish the master program?
 - An online portfolio might increase your visibility!
 - Perhaps add your course project to such portfolio?
 - [Demo](#)
-

Bibliography

- [1] H. Wickham, M. Çetinkaya-Rundel, and G. Grolemund, “R for Data Science (2e).” [Online]. Available: <https://r4ds.hadley.nz/>